

HUJI Research Monitor

Agriculture & Food · Weekly Digest · Week 29 · July 13, 2026

EXECUTIVE SUMMARY

This week's digest features two innovations with significant potential across the FoodTech and AgriTech sectors. A groundbreaking development in cultivated meat production addresses the industry's primary cost barrier by drastically reducing growth factor consumption through novel cellulose scaffolds. Complementary research offers a strategy to enhance the quality and shelf-life of traditional meat products, specifically targeting oxidation in salted beef through precision animal nutrition. These technologies offer pathways to more sustainable and higher-quality food systems.

#1

Growth factors-infused cellulose scaffolds support cost-efficient proliferation and differentiation of bovine stem cells for cultivated meat.

35/50

Main Researcher: Sharon Schlesinger

[FoodTech] [AgriTech] [Materials] [Synthetic Biology]

"Novel cellulose scaffolds drastically cut cultivated meat production costs by reducing expensive growth factor use tenfold."

WHY NOW

The cultivated meat industry is poised for rapid growth, with production costs being the key hurdle to scalability and market adoption. This technology provides a critical, near-term solution to make cultivated protein economically viable for mass markets.

COMMERCIAL ANGLE

This technology addresses the single largest cost barrier in cultivated meat production by significantly reducing the consumption of expensive growth factors through localized scaffold delivery.

<https://europepmc.org/article/MED/42182043>

#2

Vitamin E supplementation moderates salt-induced oxidation status in beef.

23/50

Main Researcher: Oren Tirosh

[AgriTech] [FoodTech]

"Vitamin E supplementation in livestock extends shelf-life and prevents discoloration in premium salted beef products."

WHY NOW

As consumers increasingly demand high-quality, natural food products with extended freshness, this offers a direct method for producers to enhance product appeal and reduce waste in a premium market segment.

COMMERCIAL ANGLE

Development of specialized livestock feed additives or precision nutrition protocols to extend the shelf-life and visual appeal of premium salted beef products.

<https://pubmed.ncbi.nlm.nih.gov/42229083/>